Enhancing Credit Risk Prediction through Co-trained Hybrid Quantum Transfer Learning Model

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***Abstract*—In this work, we explore the potential of a co-trained hybrid quantum-classical transfer learning model in enhancing credit risk classification performance. The proposed Quantum Transfer Learning (QTL) framework integrates a classical neural network with a parameterized quantum circuit, training both components end-to-end to enable synergistic learning. For baseline comparison, we evaluated a standalone Quantum Neural Network (QNN) trained directly on raw credit risk data and a conventional classical neural network. Although the standalone QNN demonstrates theoretical feasibility, it suffers from poor recall and unstable convergence due to the lack of feature preprocessing and high computational demands. Experimental evaluations on a real-world normalized credit risk dataset show that the proposed co-trained QTL model significantly outperforms the baselines. In particular, it achieves a higher area under the ROC curve (AUC = 0.9134) and improved recall (0.73) for the minority class (defaults), underscoring its effectiveness in addressing class imbalance while remaining feasible under current quantum hardware constraints.**

***Index Terms*—Quantum Machine Learning, Transfer Learning, Feedforward Neural Network, Quantum Neural Network**

I. Introduction

Transfer learning refers to the process of utilizing a model that has been pre-trained on a specific task and adapting it either wholly or partially to solve a related but distinct task. This technique is particularly useful in scenarios where labeled data are limited, allowing models trained on larger, similar datasets to provide a strong starting point. Most of the literature uses Convolutional Neural Network (CNN) by adding new layers and freezing the trained portion, then allowing the model to train until the overall system operates satisfactorily in terms of accuracy [10]. However, transfer learning is not just for image-related application tasks [24]. Although traditionally associated with computer vision, transfer learning has demonstrated effectiveness across a broad spectrum of machine learning architectures, including Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Graph Neural Networks (GNNs) and even Large Language Models (LLMs), with applications spanning sentiment analysis, time series forecasting, biomedical research and drug discovery [12], [17]–[19]. Beyond improving model generalization in low-data regimes, transfer learning contributes significantly to computational efficiency by reducing the need to train models from scratch. This leads to a reduction in overall training time and associated carbon emissions, making it an environmentally conscious approach in modern AI pipelines [20].

Risks are an intrinsic part of financial systems that influence decisions at the individual and institutional levels. Therefore, an accurate assessment of credit risk is essential to maintain financial health, minimize loan defaults and comply with regulatory standards [26]. The shortcomings of traditional risk modeling were starkly highlighted during the 2008 global financial crisis, underscoring the need for more robust, transparent and predictive analytical systems. In response, modern financial risk management increasingly incorporates machine learning [27] [28] and more recently, quantum-enhanced models [29] that aim to capture complex data dependencies and improve minority class prediction, particularly in imbalanced credit risk [30] datasets. Classical credit risk models have evolved from simple heuristic methods to sophisticated machine learning techniques where logistic regression remains a benchmark due to its interpretability [16]. Recent advancements include ensemble methods [31] like Random Forests and also Deep Learning (DL) approaches, which have shown improved predictive power but at the cost of interpretability. Despite these advances, current credit risk models still struggle with handling non-linear relationships in high-dimensional data and handling minimum class recall [32].

This paper addresses the following core research challenges

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and contributions:

- **Co-trained Quantum Transfer Learning for Credit Risk Assessment:** Demonstrates how a fully co-trained hybrid quantum-classical model can be effectively applied to real-world financial data sets to improve predictive performance under unbalanced data conditions.
- **Improved Minority Class Detection:** Significantly boosts recall for the minority or defaulter class (recall = 0.73), addressing a critical limitation in traditional credit scoring models by learning more sensitive and quantum compatible feature representations.
- **Quantitative Performance Gains:** Achieves improved area under the ROC curve (AUC = 0.9134) and high classification accuracy, outperforming both the classical baseline and the standalone QNN in separating default from non-defaulter classes.
- **End-to-End Optimization and Convergence Behavior:** Leverages simultaneous training of classical and quantum layers to enable mutual adaptation. Although convergence is more dynamic, it leads to better generalization for the minority class compared to fixed feature models.
- **Feasibility under Quantum Hardware Constraints:** Demonstrates that quantum-assisted learning can be effectively integrated into financial decision-making pipelines using current quantum simulators, despite moderate computational overhead from co-training.

II. Background

Quantum Machine Learning (QML) has emerged as a promising interdisciplinary field that uses quantum computing principles to enhance machine learning algorithms. Recent advancements in QML include QNN's [3], Variational Quantum Algorithms (VQA) like QAOA and VQE [4], quantum kernel methods and quantum feature maps [5] and of course QTL approaches [13]. Notable progress has been made in the implementation of quantum versions of classical algorithms that demonstrate a potential quantum advantage for specific problems [6]. However, QML faces significant challenges including hardware limitations (qubit count, coherence time and noise susceptibility), generalization gaps between theoretical proposals and practical implementations and limited benchmarking studies on real-world datasets [7] [8]. Practical quantum computing still faces many hurdles before it can solve real-world problems, although it looks promising on paper [1]. Stable calculations are hard to maintain for long periods of time due to quantum decoherence and Noisy Intermediate Scale Quantum (NISQ) defects [2]. Data encoding inefficiency represents a critical barrier to the practical deployment of quantum computing systems.

Classical transfer learning has emerged as a pivotal technique in machine learning where the knowledge gained from solving one problem is applied to a different but related domain [21] [23]. This approach has become increasingly important as it addresses the challenge of limited labeled data by leveraging pre-trained models on large datasets to improve performance on specialized tasks with fewer samples [22]. Recent advances have focused on domain adaptation techniques and feature representation transfer that maintain performance despite distribution shifts between source and target domains.

Quantum Transfer Learning presents a promising solution to the challenge of data encoding inefficiency in quantum machine learning. Using pre-trained classical neural networks for initial feature extraction, QTL effectively reduces the dimensionality of input data prior to quantum processing [11]. This approach ensures that only a selected subset of computationally significant features, those most suitable for quantum representation, are encoded into quantum states. Consequently, the computational burden on quantum hardware is substantially minimized, allowing efficient resource utilization without sacrificing the potential benefits of quantum computation [10]. This hybrid framework not only addresses the bottleneck of full quantum model implementation, but also enables practical, near-term deployment of quantum-enhanced learning methods across various fields such as finance, healthcare and combinatorial optimizations.

Credit risk assessment remains a data-intensive and analytically demanding task in which financial institutions depend on predictive models to evaluate the likelihood of loan default [14]. While conventional machine learning techniques have shown utility in this domain, they often do not capture higher-order interactions, latent correlations and nonlinear dependencies that are inherent in complex financial data sets [15]. Quantum Transfer Learning, which combines classical neural networks for initial feature extraction with quantum circuits for advanced pattern recognition, introduces a powerful hybrid methodology for overcoming these limitations. Furthermore, QTL helps mitigate the class imbalance issues commonly observed in credit datasets, leading to more balanced and reliable predictions. However, challenges remain in determining optimal quantum-classical boundaries and mitigating the effects of noise on current quantum hardware [9].

III. Methodology: Co-trained Quantum Transfer Learning Architecture

Traditional hybrid quantum-classical models often employ a frozen classical feature extractor, where the classical neural network is pre-trained on a large-scale dataset (e.g., ImageNet or domain-specific financial data) and its parameters remain unchanged during the training of the quantum component. This approach limits the adaptability of the model to new tasks or datasets. In contrast, this model adopts a co-training strategy, where both the classical and quantum components are trained simultaneously. This end-to-end training allows the classical feature extractor to adapt its representations in response to feedback from the quantum layer, enabling a more synergistic learning process.

A. Model Architecture

The Co-trained Hybrid Quantum Transfer Learning architecture (Fig. 1) comprises of two primary components:

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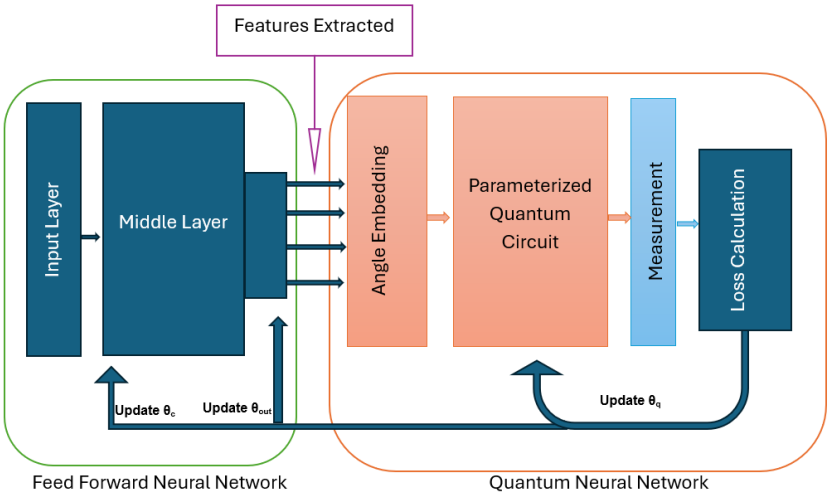

Fig. 1: Co-trained Hybrid Quantum Transfer Learning Model

- **Classical Feature Extractor:** A neural network that processes input data and extracts relevant features.
- **Quantum Circuit:** A parameterized quantum circuit that receives the features of the classical extractor and performs further transformations to aid in classification.

During training, the error is propagated back through both classical and quantum components, allowing joint optimization. This co-training approach aims to enhance the model's ability to capture complex patterns in the data, particularly beneficial in tasks with class imbalance, such as credit risk assessment.

B. Training Procedure

The training process involves the following steps:

1) Initialize the parameters of both the classical neural network and the quantum circuit.
2) For each batch of training data
   a) Pass the data through the classical feature extractor to obtain feature representations.
   b) Encode these features into quantum states and process them through the parameterized quantum circuit.
   c) Measure the output of the quantum circuit to obtain predictions.
   d) Calculate the loss between the predictions and the true labels.
   e) Back-propagate the error through the quantum circuit and the classical network to update their parameters.

Co-training enables simultaneous optimization of classical and quantum components, enhancing adaptability to new data distributions and boosting minority class prediction.

IV. Mathematical Formulation

Unlike conventional neural networks, the classical feature extractor network does not include a task-specific output layer, such as softmax or the sigmoid layer. Instead, it terminates at the 4 neuron feature extraction layer. These learned features are then passed as input to the quantum circuit, where each value is encoded on a separate qubit using rotational gates, such as $\mathrm{R}_Y(\theta_i)$ for each qubit q_i. The quantum circuit consists of an embedding layer, an entanglement layer (e.g., CNOT gates) and a series of trainable parameterized rotation gates that form a variational quantum block. After quantum processing, the circuit performs expectation measurements on selected qubits. The result is passed through a classical linear output layer to generate the final prediction.

This design allows the classical model to compress high-dimensional input into a low-dimensional quantum representable form, reducing the quantum hardware requirements while leveraging quantum computation for higher-order pattern recognition.

Let the input feature vector be denoted as:

$$\mathbf{x} = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^n \tag{1}$$

where n is the number of input features (e.g., $n = 23$ for the credit risk dataset). The classical feature extractor is a feedforward neural network represented as a function:

$$\mathbf{z} = f_{\text{classical}}(\mathbf{x}; \theta_c) \tag{2}$$

where θ_c are the pre-trained parameters of the classical network and $\mathbf{z} \in \mathbb{R}^d$ is the low-dimensional feature vector extracted from the final hidden layer (with $d = 4$).

These extracted features $\mathbf{z} = [z_1, z_2, z_3, z_4]$ are then passed to the quantum circuit. Each component $z_i \in [-1, 1]$ is encoded into a single qubit using angle encoding through a parameterized rotational gate, typically the R_Y gate:

$$|\psi_i\rangle = \mathrm{R}_Y(z_i)\,|0\rangle = \cos\left(\frac{z_i}{2}\right)|0\rangle + \sin\left(\frac{z_i}{2}\right)|1\rangle \tag{3}$$

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The full quantum state of the 4-qubit system after encoding is as follows:

$$|\psi\rangle = \mathrm{R}_Y(z_1) \otimes \mathrm{R}_Y(z_2) \otimes \mathrm{R}_Y(z_3) \otimes \mathrm{R}_Y(z_4) \, |0000\rangle \quad (4)$$

This quantum state is then processed by a variational quantum circuit composed of entanglement gates and trainable parameterized unitaries $U(\theta_q)$:

$$|\phi\rangle = U(\theta_q) \, |\psi\rangle \quad (5)$$

where θ_q are the parameters of the quantum circuit. In existing models, only the quantum parameters θ_q are optimized during training, while the classical model remains frozen. Finally, a measurement is performed on the computational basis (e.g. Z-basis) and the output is used for classification.

$$\hat{y} = \sigma\left(\mathbf{w}^\top \langle\phi|Z|\phi\rangle + b\right) \quad (6)$$

where $\sigma(\cdot)$ is the sigmoid activation function and $\mathbf{w}, b$ are the weights and bias of the final classical output layer that maps the quantum circuit output to a probability score. Unlike existing models, the parameters θ_c of the classical network are not frozen, but are instead updated during training to improve synergy with the quantum circuit. This learned feature vector $\mathbf{z}$ is passed to a 4-qubit variational quantum circuit. Each feature z_i is encoded in the quantum state using angle encoding through single-qubit $RY(z_i)$ rotations as mentioned in Eq. (3). This encoded state is processed by a trainable variational circuit and the model is trained using binary cross-entropy loss:

$$\mathcal{L}(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y}), \quad (7)$$

and the loss gradient is back-propagated through the output layer, quantum circuit and classical network. As a result, all parameters θ_c, θ_q and θ_{out} are updated during optimization.

V. Experimental setup

This section describes the dataset used for model evaluation, the experimental setup and training workflow for circuit construction and optimization.

A. Credit Risk Dataset

In this study, the Kaggle credit risk dataset is used, which contains 24 features and 32,580 data instances [35]. Those who defaults come under class 1, which is minority class in the dataset. Initially, a comprehensive exploratory data analysis (EDA) was performed and the dataset was normalized using Min-Max scaling. Initial analysis revealed a class imbalance within the dataset, prompting the application of Synthetic Minority Over-sampling Technique (SMOTE) to address this issue. The experimental results indicated that the unbalanced dataset yielded higher accuracy, attributed to overfitting in the synthetic samples generated by SMOTE. Consequently, the cleaned, normalized and unbalanced dataset was selected for training the proposed quantum-classical hybrid model.

B. Quantum Neural Network

To evaluate the standalone QNN baseline, we designed a minimal implementation using a four-qubit variational quantum circuit trained on a stratified subset of the credit risk dataset. Due to the significant computational overhead associated with parameter shift gradient calculations and single-sample quantum circuit evaluations, we limited the training data to 1,000 samples. Stratified sampling was applied to preserve class balance during this reduction. The classical features were linearly projected into a 4-dimensional latent space before being angle-encoded into the quantum circuit. Training was performed on a TPU on Colab environment using the PennyLane framework and the model was evaluated across 10 epochs to assess convergence and generalization.

C. Hybrid Quantum Transfer Learning

The overall hybrid model is implemented using PennyLane's `TorchLayer` interface, which allows the quantum circuit to be embedded in a PyTorch pipeline. The forward pass of the combined model consists of: feeding the output into the quantum layer and passing the result through a final `nn.Linear(1,1)` layer followed by a sigmoid activation for binary classification. Training is performed using the Binary Cross Entropy (BCE) loss function and optimization is conducted via the Adam optimizer.

So we can consolidate the experimental structure as :

- **Dataset:** Credit risk dataset with normalized features and binary labels (loan default prediction)
- **Split:** 80% training, 20% testing with stratification
- **Feature Extractor:** 3-layer Multi Layer Perceptro with ReLU and Tanh activations
- **Quantum Circuit:** 4-qubit variational circuit (PennyLane, angle encoding + entanglement)
- **Loss Function:** Binary Cross Entropy
- **Optimizer:** Adam

In this model, the classical network is not frozen; instead, it is co-trained and fine-tuned alongside the quantum component. Thus, the model learns both classical and quantum weights together. This is a valid learning strategy, especially if you want to allow mutual adaptation between layers. This gives you the best of both worlds: Knowledge reuse (transfer learning) and flexibility (fine-tuning). All parameters of the classical neural network (θ_c), the quantum circuit (θ_q) and the output layer (θ_{out}) are trainable (Eq. (7)) and optimized with a learning rate of 0.01.

Although GPU acceleration is commonly used to speed up classical deep learning tasks, our experiments employ the PennyLane default.qubit backend, which is a CPU-based quantum simulator. As such, quantum circuit simulation, particularly parameter shift gradient computation, remains the dominant computational bottleneck, regardless of GPU availability [25], [33], [34]. The model is trained for 10 epochs and the performance is monitored based on the loss and accuracy of the test.

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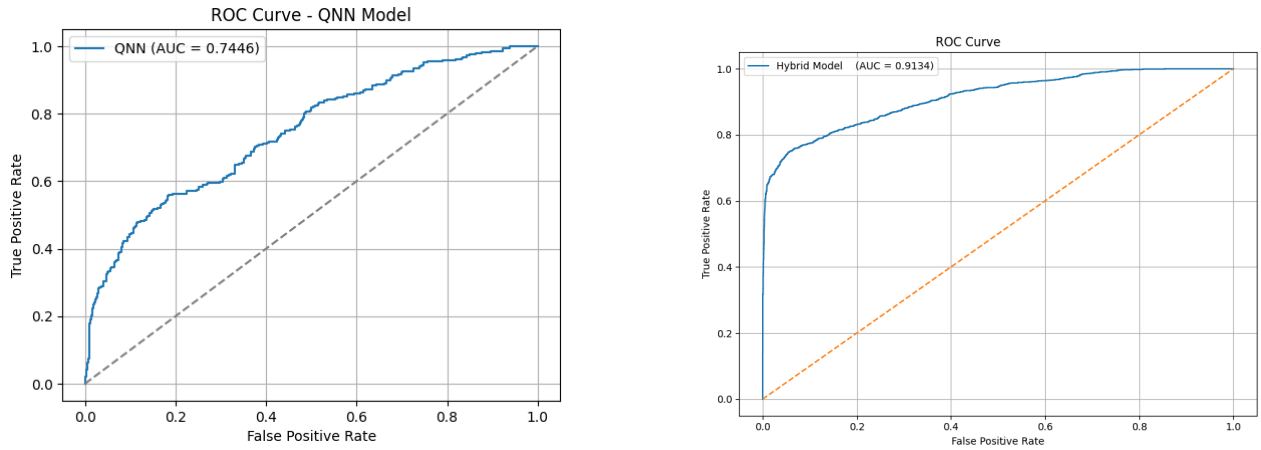
TABLE I: Performance Comparison of Standalone QNN and Hybrid Co-trained QTL Model on Credit Risk Dataset

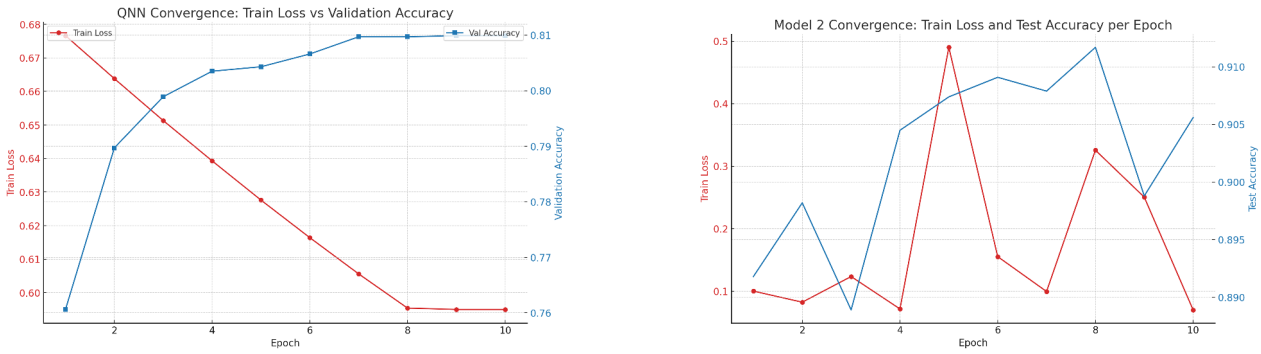
Metric	QNN (Standalone)	Hybrid Co-trained QTL Model
Accuracy	0.8097	0.9117
AUC Score	0.7446	0.9134
Precision (Class 0)	0.81	0.93
Recall (Class 0)	0.99	0.96
F1-Score (Class 0)	0.89	0.94
Precision (Class 1)	0.82	0.82
Recall (Class 1)	0.16	0.73
F1-Score (Class 1)	0.27	0.77



(a) ROC Curve for the QNN Model

(b) ROC Curve for Hybrid Co-trained QTL Model

Fig. 2: ROC Curves of QNN and Hybrid Co-trained QTL Model for Credit Risk Classification



(a) Standalone QNN Convergence

(b) Hybrid QTL Co-trained Model Convergence

Fig. 3: Convergence Curves of Standalone QNN and Hybrid QTL Co-trained model

VI. Results

Table I provides a detailed comparative analysis of the evaluation metrics for the standalone QNN and the hybrid co-trained QTL model. A notable improvement is observed in both accuracy and recall when comparing the QTL-based model to the standalone QNN, indicating the effectiveness of quantum transfer learning in enhancing predictive performance, particularly for the classification of imbalanced credit risks. The ROC/AUC associated with each model is specified in Fig. 2.

A. Performance Evaluation

During training, model demonstrated steady convergence, reaching its optimal performance at epoch 8 with a test accuracy of 91.17%. The model preserves strong performance of the majority class while increasing the recall of the minority class from 0.16 to 0.73, enhancing the sensitivity to defaulters in the assessment of credit risk. The area under the ROC curve (AUC) is 0.9134, this superior AUC demonstrates the model's increased reliability in ranking defaulters above non-defaulters based on predicted probability scores. The f1 score

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for class 1 improves from 0.27 to 0.77, underscoring the utility of quantum transfer learning to capture complex and high-dimensional dependencies. Nevertheless, this model incurs a higher computational cost as a result of the joint training of the classical and quantum layers.

B. Observation on Convergence Variability And Training time

The standalone QNN showed a more stable convergence profile, while the hybrid QTL model (Fig. 3) exhibited greater fluctuations in training loss and test accuracy. This variability stems from non-convex circuit training, simultaneous classical-quantum parameter updates and high-variance gradients from the parameter-shift rule. Despite these oscillations, the hybrid model demonstrated an overall upward performance trend, confirming effective learning and highlighting the need for careful optimization strategies. Table II shows that the standalone QNN requires nearly twice the training time compared to the hybrid co-trained QTL model, making it not only computationally more efficient but also a time-efficient approach.

TABLE II: Approximate Training Time of models

Model	**Training Time (Approx.)**
Standalone QNN	9 hours
Hybrid QTL Model	3 hours

Conclusion

This study demonstrates the practical benefits of quantum transfer learning for credit risk assessment. Although the classical model performs well on the majority class, it lacks sensitivity to minority class defaulters. The standalone QNN, although conceptually appealing, suffers from poor recall and high training cost. In contrast, the hybrid co-trained QTL model shows marked improvements in recall and AUC, achieving the best performance through co-training. However, this comes at the expense of increased computational overhead. These findings underscore that hybrid architectures offer a meaningful path forward, balancing performance gains with the limitations of current quantum technologies.

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